

NUUK GODTHAAB, Greenland

WMO: 042500

Lat: 64.167N

Lon: 51.744W

Elev: 276

StdP: 14.55

Time Zone: -3.00 (GRL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.3	1.5	-10.5	3.2	1.6	-6.3	4.0	4.8	49.9	30.9	44.1	26.1	15.5	0	0.948

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	59.4	49.7	56.4	48.6	53.7	47.3	51.3	56.8	49.8	54.5	48.3	52.6	9.2	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
48.5	50.9	52.1	46.7	47.7	50.5	45.3	45.2	49.3	21.1	57.1	20.2	54.8	19.4	52.7	60.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 44.3	35.9	29.5	-3.1	66.6	5.5	3.8	-7.0	69.3	-10.3	71.5	-13.4	73.7	-17.4	76.4		
(5)			-3.7	55.3	5.4	2.3	-7.6	57.0	-10.8	58.3	-13.8	59.6	-17.7	61.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	31.3	19.4	17.8	18.8	27.5	35.1	42.1	46.4	45.7	39.9	32.5	26.7	22.4
	DBStd	12.43	8.85	10.17	10.48	7.56	5.75	5.07	4.62	4.28	4.10	6.03	7.14	8.16
	HDD50	6878	949	903	967	676	462	241	130	146	305	543	700	856
	HDD65	12313	1414	1323	1432	1126	927	686	575	598	754	1007	1150	1321
	CDD50	41	0	0	0	0	1	5	20	13	1	1	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	12.6	14.4	14.2	14.1	12.9	11.4	10.7	10.3	10.7	12.0	12.4	14.4	14.3
Precipitation	PrecAvg	29.2	1.8	1.8	1.8	1.8	2.1	2.0	2.8	3.5	3.4	3.0	2.7	2.1
	PrecMax	45.1	7.9	8.7	6.0	5.6	5.8	6.6	9.0	8.9	8.7	7.1	8.3	7.9
	PrecMin	18.0	0.1	0.1	0.1	0.2	0.1	0.0	0.1	0.0	0.6	0.4	0.3	0.2
	PrecStd	6.1	1.4	1.6	1.3	1.3	1.5	1.5	1.9	2.2	1.8	1.7	1.9	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.7	47.9	46.7	49.1	56.3	62.2	63.4	62.5	55.8	52.7	50.1	46.0
		MCWB	36.1	37.6	39.3	39.4	46.1	50.5	51.5	52.5	46.4	42.7	41.7	37.4
	2%	DB	38.8	41.3	39.6	42.9	49.6	57.0	60.0	57.4	49.8	46.2	43.5	40.3
		MCWB	33.2	33.9	34.3	36.9	42.5	48.3	50.1	49.7	44.2	39.6	37.8	35.1
	5%	DB	35.5	36.0	35.9	39.3	46.0	53.4	57.0	54.2	47.2	42.5	39.1	36.8
		MCWB	31.3	31.1	31.8	35.3	40.4	46.3	49.2	48.5	42.9	38.1	35.0	33.1
	10%	DB	32.0	31.7	33.1	36.7	42.7	50.1	54.0	51.8	45.1	39.6	35.8	33.8
		MCWB	28.9	28.8	30.1	33.8	38.8	44.7	47.7	47.3	41.7	36.3	33.0	31.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.6	39.2	39.6	41.0	46.8	51.5	53.7	54.5	49.2	45.0	43.4	39.6
		MCDB	42.7	43.8	44.6	45.0	54.2	58.3	59.9	59.8	53.8	49.2	47.3	42.9
	2%	WB	34.3	34.8	35.3	37.5	43.3	49.1	51.4	51.2	46.0	41.1	38.8	36.1
		MCDB	37.4	40.1	38.2	41.0	48.2	55.5	57.5	55.4	48.2	44.2	42.6	39.4
	5%	WB	31.4	31.4	32.5	35.7	41.0	46.8	50.0	49.5	43.9	38.6	35.5	33.5
		MCDB	34.5	35.2	35.4	38.6	44.6	52.1	55.5	52.8	45.9	41.5	38.4	36.1
	10%	WB	28.7	28.7	29.9	33.9	39.1	45.0	48.5	48.0	42.2	36.6	33.1	31.0
		MCDB	32.1	32.1	33.0	36.5	42.1	49.4	53.4	51.1	44.3	39.3	35.3	33.8
Mean Daily Temperature Range	5% DB	MDBR	7.7	8.1	8.4	8.1	8.7	11.1	11.6	9.9	7.5	6.6	6.9	7.4
		MCDBR	12.7	15.0	13.2	12.0	14.6	17.1	16.8	14.4	11.0	11.2	12.3	11.5
		MCWBR	9.7	10.2	9.2	7.9	8.8	10.0	9.2	8.2	6.9	7.4	8.9	8.6
	5% WB	MCDBR	12.5	14.7	12.7	11.3	13.3	15.7	14.9	13.0	10.2	10.1	12.0	11.1
		MCWBR	10.1	10.8	9.6	7.9	8.8	9.9	9.1	8.1	7.3	7.5	9.4	8.9
Clear-Sky Solar Irradiance	taub	0.197	0.245	0.274	0.317	0.340	0.345	0.343	0.345	0.306	0.268	N/A	0.350	
	taud	1.950	2.294	2.360	2.301	2.353	2.424	2.475	2.489	2.523	2.375	N/A	2.904	
	Ebn at Noon	118	224	264	273	276	277	274	263	252	210	N/A	80	
	Edh at Noon	9	16	24	32	33	32	30	27	21	15	N/A	5	
All-Sky Solar Radiation	RadAvg	39	195	637	1210	1678	1817	1657	1215	720	322	71	17	
	RadStd	4	26	65	121	129	141	172	108	84	33	15	2	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.88	+2.93	N/A	+2.02	N/A	N/A	N/A	N/A	+20	N/A
(47) Regional (3 neighbors)		+0.88	+2.93	N/A	+2.02	N/A	N/A	N/A	N/A	+20	N/A

Nomenclature: See separate page